

An improved AUSM-family scheme with robustness and accuracy for all Mach number flows



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ABSTRACT

With the increasing popularity of Computational Fluid Dynamics (CFD), the reliability of numerical scheme becomes prominent. The work presents a newly improved scheme more reliable in all Mach number regimes to circumvent some typical symptoms of the previous AUSM-family schemes observed in hypersonic and very low speeds. This scheme is facilitated by reconstructing pressure diffusion term in mass flux, velocity diffusion term in pressure flux and numerical sound speed. Then, a variety of benchmark test cases are selected to systematically assess the effects of these key ingredients and investigate the additional features in terms of robustness and accuracy. The proposed scheme attains stronger shock robustness against carbuncle instability, better low-speed accuracy and higher resolution of oblique shocks compared with many existing upwind schemes. Moreover, it can exactly resolve contact discontinuity, preserve positivity, damp numerical overshoots and avert the global cut-off strategy. Numerical results for a wide spectrum of Mach numbers indicate its potential and reliable application to all Mach number flows.

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1. Introduction

Nowadays, Computational Fluid Dynamics (CFD) has become an important analysis and design technique in aeronautics and astronautics. With the increasingly widespread use of CFD, the reliability of computed solutions is drawing more and more attention in practical engineering. As an essential component of CFD, upwind schemes have prevailed and matured in simulating compressible flows. Reliability in numerical scheme signifies robustness and accuracy. However, many investigations have demonstrated that upwind schemes face some difficulties in hypersonic [1–9] and very low speeds [10–19]. Consequently, their reliability in the computation of all Mach number flows are reduced. Thus, further development of advanced numerical scheme with high level of robustness and accuracy is very inspiring.

Upwind schemes are generally divided into two categories: Flux Difference Splitting (FDS) and Flux Vector Splitting (FVS). Based on the wave structure, FDS methods can be classified as complete and incomplete approximate Riemann solvers. The complete approximate Riemann solvers have three waves involving two nonlinear waves and a contact discontinuity, which are same as the exact Riemann solver. Representatives of complete solvers are Roe [20], HLLC [21] and Osher [22] schemes. These methods have been widely prevalent and acceptable due to the outstanding capability of resolving contact

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discontinuity, boundary layer and constant shock. Nevertheless, complete solvers encounter the disastrous failing known as carbuncle phenomenon [1] or shock instability for hypersonic vehicles, especially blunt body shape. Many approaches have been proposed to cure carbuncle instability, the most typical of which is entropy correction in Roe scheme. However, the entropy correction with better performance (e.g. Harten-Yee's entropy fix [23]) needs to set artificial empirical parameters, and its shock robustness strongly depends on the choice of these parameters. It may also smear boundary-layer resolution [2]. The incomplete approximate Riemann solvers adopt coarser wave model lack of linear wave, leading to smearing contact discontinuity and viscous resolution. The great merit of incomplete solvers is very robust against carbuncle instability. They are capable of capturing strong shocks with high efficiency. Incomplete solvers contain two-wave approximate Riemann solver (HLL [24]) and one-wave Riemann solver (Rusanov [25]). Furthermore, the FVS methods have lower algorithm complexity and computational effort without the eigenstructure requirement. FVS methods mainly include van Leer's FVS scheme [26] and Steger-Warming scheme [27]. Similar to the incomplete FDS schemes, FVS methods have good shock stability but lack contact discontinuity. Combining the advantages of FVS and FDS, Liou and Steffen [28] developed AUSM (Advection Upstream Splitting Method) through splitting inviscid flux into convective and pressure systems. It effectively inherits the robustness of FVS in shock wave and the accuracy of FDS in contact discontinuity. Due to its excellent performance with low dissipation and shock stability, this method has gradually gained acclaim and continued to develop many new variants (e.g. AUSMDV [29], AUSM+ [30], AUSMPW+ [31], etc.). Nevertheless, there are still some issues in AUSM-family schemes. For instance, AUSM+ is vulnerable to pressure wiggles near the wall and numerical overshoots behind strong shock waves [18,31]. AUSM+ is regarded as slight carbuncle-prone scheme [3,6]. Besides, shock oscillations may be caused by these AUSM-family schemes if the computational mesh has poor quality in shock region.

Upwind schemes mentioned above are suitable for compressible flow, but encounter difficult convergence and deteriorated accuracy [10,18] when solving nearly incompressible flows or very low-speed flows. To address such shortcomings, the classic preconditioning techniques [11–13] have been widely applied over the past years. On the one hand, preconditioned methods effectively improve low-speed accuracy and mitigate the stiffness. On the other hand, they yield the following drawbacks: global cut-off problem [11], reformulation of characteristic boundary conditions and sacrifice of time accuracy. Meanwhile, recent numerous efforts have extended the existing codes of upwind schemes to simulate very low speeds, accomplishing all-speed schemes. A series of the improved Roe-type schemes [14–16] have been proposed without low-speed accuracy problem and global cut-off problem. Unfortunately, they are still afflicted by carbuncle instability unless the corresponding high-Mach enhancement is incorporated. Furthermore, many researchers have made contributions to low-Mach number extensions of AUSM-family schemes: Edwards [17] proposed LDFSS towards unified CFD simulations; Liou [18] developed AUSM+UP via introducing pressure and velocity diffusion terms; Shima et al. [19] designed SLAU with low dissipation and no tunable parameters; Kitamura et al. [32] further put forward SLAU2, AUSM+UP2, LDFSS2001-2 by a new pressure flux. SLAU2 improves the robustness of SLAU, and can be considered as a good candidate across a wide range of Mach number. These all-speed AUSM-family schemes are very attractive with high resolution. However, there are some deficiencies in the respective constructs. SLAU2 or SLAU is still prone to shock anomalies for some strong shocks [6,33] as can occur when cells have large aspect ratio. AUSM+UP inevitably takes the global cut-off strategy to ensure computational stability. AUSM+UP allows an extremely small time step in low Mach number limit when not increasing the cut-off reference Mach number [34]. The incremental reference Mach number in AUSM+UP causes more numerical viscosity at low speeds [35,36]. It is also observed that AUSM+UP is not robust enough against shock anomalies and predict the poor heating distributions in some multi-dimensional hypersonic atmospheric re-entry vehicles.

Thus, the motivation of the present study is to enhance the robustness and accuracy of AUSM-family schemes, accomplishing a newly improved version more reliable to simulate all Mach number flows. The proposed scheme called AUSM+M is evolved by three key ingredients, including pressure diffusion term, velocity diffusion term and numerical sound speed. AUSM+M is very easy to implement with the negligible computational expense. One objective of the current effort is to make clear the effects of these key ingredients and the additional features of AUSM+M. The tremendous numerical results underline its excellent performance for all speeds. The rest of the paper is organized as follows: Section 2 provides governing equations and AUSM-family schemes; Section 3 develops the improved AUSM-family scheme with robustness and accuracy; Section 4 performs the basic features of the proposed scheme; Section 5 carries out numerical cases; Section 6 gives the concluding remarks.

2. Governing equations and AUSM-family schemes

2.1. Governing equations

Consider two-dimensional compressible Euler equations:

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F} + \partial_y \mathbf{G} = 0 \quad (1)$$

$$\mathbf{Q} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho e \end{bmatrix}, \mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uh \end{bmatrix}, \mathbf{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vh \end{bmatrix} \quad (2)$$

where $\mathbf{Q}$ is conservative state vector and $\mathbf{F}$, $\mathbf{G}$ are flux vectors. ρ , $\mathbf{u}=(u, v)$ and p are density, velocity and pressure. Using the perfect gas law, the speed of sound a , specific total energy e and total enthalpy h are obtained as

$$a = \sqrt{\frac{\gamma p}{\rho}}, e = \frac{p}{(\gamma - 1)\rho} + \frac{\|\mathbf{u}\|^2}{2}, h = e + \frac{p}{\rho} \quad (3)$$

with the specific heat ratio $\gamma = 1.4$. This hyperbolic system is then solved numerically via conservative method:

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n - \frac{\Delta t}{\Delta x} [\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2}] - \frac{\Delta t}{\Delta y} [\mathbf{G}_{j+1/2} - \mathbf{G}_{j-1/2}] \quad (4)$$

with the numerical flux $\mathbf{F}_{i+1/2}$ and $\mathbf{G}_{j+1/2}$.

2.2. AUSM-family schemes

Based on the rotational invariance property of Euler equations [37], finite-volume discretization demands a normal numerical flux across the interface. The inviscid flux projected in an arbitrary unit vector $\mathbf{n}=(n_x, n_y)$ is $\mathbf{F}_{1/2}=(\mathbf{F}, \mathbf{G}) \cdot \mathbf{n}=\mathbf{F}n_x + \mathbf{G}n_y$. The numerical flux calculated by AUSM-family schemes is generally expressed as follows:

$$\mathbf{F}_{1/2} = \frac{\dot{m} + |\dot{m}|}{2} \Phi_L + \frac{\dot{m} - |\dot{m}|}{2} \Phi_R + \mathbf{P}_{1/2}$$

$$\Phi = \begin{bmatrix} 1 \\ u \\ v \\ h \end{bmatrix}, \mathbf{P}_{1/2} = \begin{bmatrix} 0 \\ p_s \cdot n_x \\ p_s \cdot n_y \\ 0 \end{bmatrix}, \dot{m} = M_{1/2} c_{1/2} \begin{cases} \rho_L, & M_{1/2} \geq 0 \\ \rho_R, & M_{1/2} < 0 \end{cases} \quad (5)$$

with mass flux $\dot{m}$, pressure flux $\mathbf{P}_{1/2}$, the numerical sound speed $c_{1/2}$ and the interface Mach number $M_{1/2}$. The superscripts “+” and “-” or the subscripts “L” and “R” signify the left and right states depending on reconstruction methods. The following Mach number and pressure splitting functions are usually applied in AUSM-family formulation.

$$M_{L/R}^{\pm} = \begin{cases} \pm \frac{1}{4} (M_{L/R} \pm 1)^2 \pm \beta (M_{L/R}^2 - 1)^2, & |M_{L/R}| < 1 \\ \frac{1}{2} (M_{L/R} \pm |M_{L/R}|), & |M_{L/R}| \geq 1 \end{cases}$$

$$\psi_{L/R}^{\pm} = \begin{cases} \frac{1}{4} (M_{L/R} \pm 1)^2 (2 \mp M_{L/R}) \pm \alpha M_{L/R} (M_{L/R}^2 - 1)^2, & |M_{L/R}| < 1 \\ \frac{1}{2} (1 \pm \text{sign}(M_{L/R})), & |M_{L/R}| \geq 1 \end{cases} \quad (6)$$

$$M_{L/R} = U_{L/R}/c_{1/2}, U_{L/R} = u_{L/R} \cdot n_x + v_{L/R} \cdot n_y$$

AUSM+ [30]

Interface Mach number

$$M_{1/2} = M_L^+|_{\beta=1/8} + M_R^-|_{\beta=1/8} \quad (7)$$

Interface pressure formula

$$p_s = \psi_L^+|_{\alpha=3/16} p_L + \psi_R^-|_{\alpha=3/16} p_R \quad (8)$$

Numerical sound speed

$$c_{1/2} = \min(\tilde{c}_L, \tilde{c}_R), \tilde{c}_{L/R} = (c^*)^2 / \max(c^*, |U_{L/R}|)$$

$$h_t = \frac{\gamma RT}{\gamma - 1} + \frac{u^2 + v^2}{2} = \frac{(\gamma + 1)(c^*)^2}{2(\gamma - 1)} \quad (9)$$

AUSM+UP [18]

AUSM+UP is enhanced from AUSM+ through constructing pressure and velocity diffusion terms, and slightly modifying numerical sound speed. AUSM+UP is known to effectively simulate all Mach number flows with high resolution.

Interface Mach number with pressure diffusion term M_p

$$M_{1/2} = M_L^+|_{\beta=1/8} + M_R^-|_{\beta=1/8} + M_p$$

$$M_p = -\frac{K_p}{f_d} \max(1 - \sigma \tilde{M}^2, 0) \frac{p_R - p_L}{\rho_{1/2} c_{1/2}^2}, \rho_{1/2} = \frac{\rho_L + \rho_R}{2} \quad (10)$$

Interface pressure formula with velocity diffusion term p_u

$$p_s = \psi_L^+|_{\alpha} p_L + \psi_R^-|_{\alpha} p_R + p_u$$

$$p_u = -K_u \cdot \psi_L^+ \Big|_\alpha \cdot \psi_R^- \Big|_\alpha \cdot (\rho_L + \rho_R) (f_a c_{1/2}) (U_R - U_L)$$

$$\alpha = \frac{3}{16} (-4 + 5f_a^2) \quad (11)$$

Numerical sound speed

$$c_{1/2} = \min(\tilde{c}_L, \tilde{c}_R), \tilde{c}_L = (c^*)^2 / \max(c^*, U_L), \tilde{c}_R = (c^*)^2 / \max(c^*, -U_R)$$

$$h_t = \frac{\gamma RT}{\gamma - 1} + \frac{u^2 + v^2}{2} = \frac{(\gamma + 1)(c^*)^2}{2(\gamma - 1)} \quad (12)$$

with the other parameters

$$f_a = M_o(2 - M_o), \quad K_p = 0.25, \quad K_u = 0.75, \quad \sigma = 1.0$$

$$M_o^2 = \min[1, \max(\tilde{M}^2, M_\infty^2)], \quad \tilde{M}^2 = \frac{U_L^2 + U_R^2}{2c_{1/2}^2} \quad (13)$$

3. Development of the improved AUSM-family scheme

AUSM+ has excessive numerical viscosity in low Mach number. Heightened from AUSM+, AUSM+UP achieves the proper low-speed accuracy, but its time step is drastically restricted in low Mach number [35,36] due to the global cut-off strategy. In addition, AUSM+ and AUSM+UP are prone to carbuncle phenomenon in high Mach number [3,6]. To overcome these deficiencies, we integrate three key ingredients to develop a new AUSM-family scheme (called AUSM+M) for better robustness and accuracy. Three key ingredients involve pressure diffusion term, velocity diffusion term and numerical sound speed.

3.1. Pressure diffusion term for mass flux

It is well known that AUSM+ possesses some good attributes and works well. However, AUSM+ concerns numerical overshoots behind strong shock waves and pressure wiggles near the wall [31]. And AUSM+ suffers from the serious checkerboard modes in low Mach number limit. These remarkable deficiencies appear since there is no corresponding dissipative mechanism in the pressure field. To handle the pressure-oscillation problem and ensure pressure-velocity coupling, Liou [18] introduced the pressure diffusion term M_p in Eq. (10) into AUSM+UP. Consequently, AUSM+UP is free from pressure oscillations in all Mach number regimes. Nevertheless, this pressure diffusion term contains the local Mach number in the denominator. It causes an undesirable phenomenon, which is that AUSM+UP inevitably takes the global cut-off strategy for computational stability. AUSM+UP only allows an extremely small time step at very low speeds unless raising the cut-off reference Mach number [34]. But when increasing this reference Mach number, the more dissipation would be introduced into AUSM+UP [35,36].

Free from this defect, we develop a new form of the pressure diffusion term similar to AUSM+UP while in a different way. This improved pressure diffusion term M_p can be expressed as

$$M_p = -\frac{1}{2} (1 - f) \cdot \frac{\Delta p}{\rho_{1/2} c_{1/2}^2} \cdot (1 - g), \quad \Delta p = p_R - p_L \quad (14)$$

where g is the multidimensional pressure-based sensing function [6] defined in Eq. (25). Shown in Fig. 1, the Mach number function f is given as

$$f = \frac{1 - \cos(\pi M)}{2}, \quad M = \min(1.0, \max(|M_L|, |M_R|)) \quad (15)$$

This definition can well achieve the transition from low speed to high speed.

Here, we emphasize how Eq. (14) is constructed. On the one hand, the denominator of the improved pressure diffusion term for AUSM+M does not involve the relevant Mach number. For AUSM+UP, f_a in the denominator approaches zero as the Mach number approaches zero at low speeds. It easily triggers computational instability and needs to be restricted. As a result, AUSM+M allows a larger time step and attains better robustness than AUSM+UP for low-speed simulation in terms of stability. On the other hand, the improved pressure diffusion term is only active in subsonic regimes and stabilize low-speed simulations. In [4], Liou presented that the pressure diffusion term in mass flux causes adverse effect on stably capturing shock waves. The dissipation generated by Δp should be alleviated in the shock wave. Considering these, the factor $(1 - f)$ is used to activate the Δp term in low Mach number regimes. In addition, the factor $(1 - g)$ can check all the adjacent interfaces effectively [6] and detect strong shock waves. Then, it further reduces the dissipation of Δp in the possible shock regions to improve shock robustness.

3.2. Velocity diffusion term for pressure flux

Through mathematical transformation suggested by LDFSS [17], SLAU [19] and SLAU2 [32], the pressure flux in Eq. (8) for AUSM+ scheme can be exactly recast as:

$$p_s = \frac{p_L + p_R}{2} + (\psi_L^+ - \psi_R^-) \frac{p_L - p_R}{2} + (\psi_L^+ + \psi_R^- - 1) \frac{p_L + p_R}{2} \quad (16)$$

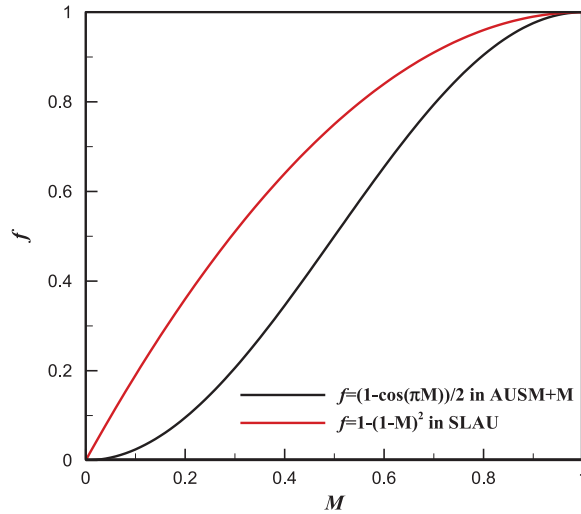


Fig. 1. The function f of SLAU and AUSM+M versus Mach number, among $f = 1 - (1 - M)^2$ for SLAU [19].

In low Mach limit, the third term of Eq. (16) can be approximated as:

$$(\psi_L^+ + \psi_R^- - 1) \frac{p_L + p_R}{2} \Rightarrow \left(\frac{3}{4} + \alpha\right) (U_L - U_R) \frac{p_L + p_R}{2c_{1/2}} \Rightarrow \left(\frac{3}{4} + \alpha\right) \frac{\rho_L + \rho_R}{2\gamma} c_{1/2} (U_L - U_R) \quad (17)$$

The parameter $\alpha = 3/16$ for AUSM+ makes Eq. (17) have the scale of acoustic speed, leading to the excessive numerical viscosity at low speeds. To improve low-speed accuracy, AUSM+UP redefines the parameter $\alpha = (-12 + 15f_a^2)/16$ to adjust Eq. (17) suitable for low-speed regimes, namely

$$\left(\frac{3}{4} + \alpha\right) \frac{\rho_L + \rho_R}{2\gamma} c_{1/2} (U_L - U_R) \Rightarrow \frac{15}{16} f_a^2 \frac{\rho_L + \rho_R}{2\gamma} c_{1/2} (U_L - U_R) \quad (18)$$

Based on Eq. (18), Eq. (17) is rescaled into the scaling of convective velocity by the Mach number function f_a . And AUSM+UP removes deteriorated accuracy at low speeds.

Eq. (17) is the error source deteriorating the low speed flow simulation. In order to make the numerical dissipation more reasonable and controllable, we use the scaling Mach number function to modify the third term of Eq. (16) following LDFSS [17] and SLAU [19]. Then, the form of the pressure term can achieve high low-speed resolution

$$p_s = \frac{p_L + p_R}{2} + (\psi_L^+ - \psi_R^-) \frac{p_L - p_R}{2} + f_o \cdot (\psi_L^+ + \psi_R^- - 1) \frac{p_L + p_R}{2} \quad (19)$$

Different from that of SLAU [19], the definition of the scaling Mach number function is as follows

$$f_o = \min(1.0, \max(f, M_\infty^2)) \quad (20)$$

where f is defined as Eq. (15).

Fig. 1 shows the scaling Mach number functions of SLAU and AUSM+M versus Mach number. As can be seen, they are monotonically increasing polynomial of Mach number in $[0, 1]$. Moreover, the value of the function f in AUSM+M (Eq. (15)) is less than that of SLAU ($f = 1 - (1 - M)^2$) at low speeds. Therefore, AUSM+M should achieve lower numerical dissipation and higher low-speed resolution than SLAU.

In addition, there are the added velocity diffusion term p_u in AUSM+UP. This velocity diffusion term is proportional to the velocity difference and possesses a notable advantage of giving smooth shock profiles. It is obtained by approaching the characteristic relation $dp \pm \rho a \cdot du = 0$ [18]. The interface pressure for $|u| < c$ is acquired via integrating the characteristic relation

$$p_s = \frac{p_L + p_R}{2} - \frac{1}{2} (\rho c)_{1/2} (u_R - u_L) \quad (21)$$

The velocity diffusion term p_u is similar to the second term of Eq. (21)

$$p_u = -K_u \psi_L^+ \psi_R^- (\rho_L + \rho_R) (f_a c_{1/2}) (U_R - U_L) \quad (22)$$

The coefficients involving ψ_L^+ and ψ_R^- can reach automatic transition between subsonic and supersonic conditions [18]. This form of the velocity diffusion term is effective in many situations, but it is gained in line with the one-dimensional characteristic relation. In some multi-dimensional hypersonic vehicles, AUSM+UP is still prone to induce shock instability. For the better shock robustness, the improved velocity diffusion term involving the transverse information is developed. To

get more consistency in pressure flux, the average density value is approximated by the combination of the average pressure value and numerical sound speed

$$\frac{(\rho_L + \rho_R)c_{1/2}}{2} \approx \frac{\gamma(p_L + p_R)}{2c_{1/2}} \quad (23)$$

The previous velocity diffusion term for AUSM+UP only contains the normal velocity difference. The current velocity diffusion is given as

$$\bar{p}_{ux} = -\frac{\gamma(p_L + p_R)}{2c_{1/2}} \psi_L^+ \psi_R^- (u_R - u_L), \bar{p}_{uy} = -\frac{\gamma(p_L + p_R)}{2c_{1/2}} \psi_L^+ \psi_R^- (v_R - v_L) \quad (24)$$

Eq. (24) is similar to Eq. (22) in one-dimension and meantime contains the transverse velocity in multi-dimension compared with Eq. (22). It is generally considered that multidimensional effect plays a key role in carbuncle phenomenon [5,6]. Lateral perturbation transfer of shock waves easily triggers shock instability. Eq. (24) provides an additional dissipation mechanism to suppress perturbation transfer and combat shock instability. However, the velocity diffusion term using Eq. (24) easily smears the shear layer. Thus, the multidimensional pressure-based sensing function [6] is introduced to detect strong shock wave and preserve shear layer. Following [6] by the authors, the pressure-based sensing function g is defined as

$$g = \frac{1 + \cos(\pi h)}{2}, h = \min_k (h_k), h_k = \min \left(\frac{p_{Lk}}{p_{Rk}}, \frac{p_{Rk}}{p_{Lk}} \right) \quad (25)$$

where the subscript k is the index for all adjacent interfaces of both cells in evaluating numerical flux. Combining Eq. (24) and Eq. (25), the improved velocity diffusion is expressed as

$$p_{ux} = -g \cdot \frac{\gamma(p_L + p_R)}{2c_{1/2}} \psi_L^+ \psi_R^- (u_R - u_L), p_{uy} = -g \cdot \frac{\gamma(p_L + p_R)}{2c_{1/2}} \psi_L^+ \psi_R^- (v_R - v_L) \quad (26)$$

Then, the improved pressure flux can be accomplished by Eq. (19) and Eq. (26)

$$\mathbf{P}_{1/2} = \begin{bmatrix} 0 \\ p_s \cdot n_x + p_{ux} \\ p_s \cdot n_y + p_{uy} \\ 0 \end{bmatrix} \quad (27)$$

with $\alpha = 3/16$.

3.3. Numerical sound speed

The numerical sound speed $c_{1/2}$ of AUSM-family schemes has the influence on numerical stability, shock robustness and physical resolution. The numerical sound speed of AUSM+ (Eq. (9)) has been widely used in many cases since it can resolve a normal shock, capture contact discontinuity and keep the conservation of total enthalpy. Even so, this numerical sound speed cannot distinguish an entropy-violating expansion shock from compression shock [18,31]. In order to get over this drawback, Liou [18] modified a new version (Eq. (12)) applied to AUSM+UP which simply adds the information of flow direction. Despite this, two definitions erase an oblique shock even with shock-aligned grids [31]. To enhance the resolution of oblique shock and remove the unphysical expansion shock, we utilize the numerical sound speed of AUSMPW+ suggested by Kim et al. [31]

$$c_{1/2} = \begin{cases} \frac{c_s^2}{\max(|U_L|, c_s)}, & \frac{1}{2}(U_L + U_R) \geq 0 \\ \frac{c_s^2}{\max(|U_R|, c_s)}, & \frac{1}{2}(U_L + U_R) < 0 \end{cases} \quad (28)$$

where c_s satisfies the Prandtl relation via an oblique shock

$$c_s = \sqrt{2(\gamma - 1)/(\gamma + 1)h_{normal}}, h_{normal} = 0.5(h_{total,L} - 0.5V_L^2 + h_{total,R} - 0.5V_R^2), V_{L/R}^2 = u_{L/R}^2 + v_{L/R}^2 \quad (29)$$

Finally, we accomplish the improved AUSM-family scheme called AUSM+M that adopts pressure diffusion term (Eq. (14)), pressure flux (Eq. (27)) and numerical sound speed (Eq. (28)). AUSM+M is very easy to implement with the negligible computational expense.

4. The basic features of the improved scheme

In this section, the effects of pressure diffusion term, velocity diffusion term and numerical sound speed are systematically assessed. Furthermore, the additional features of the improved scheme are investigated. Notice that these methods (Roe, AUSM+, AUSM+UP, SLAU and SLAU2) are chosen to demonstrate the performance of AUSM+M scheme in the following numerical examples. The reasons for this choice are as follows: (a) Roe and AUSM-type schemes are one of the most widely-used approximate Riemann solvers in simulating compressible flows; (b) The proposed AUSM+M scheme introduces some new key ingredients based on AUSM+ and AUSM+UP; (c) SLAU and SLAU2 have high resolution over wide-ranging Mach numbers.

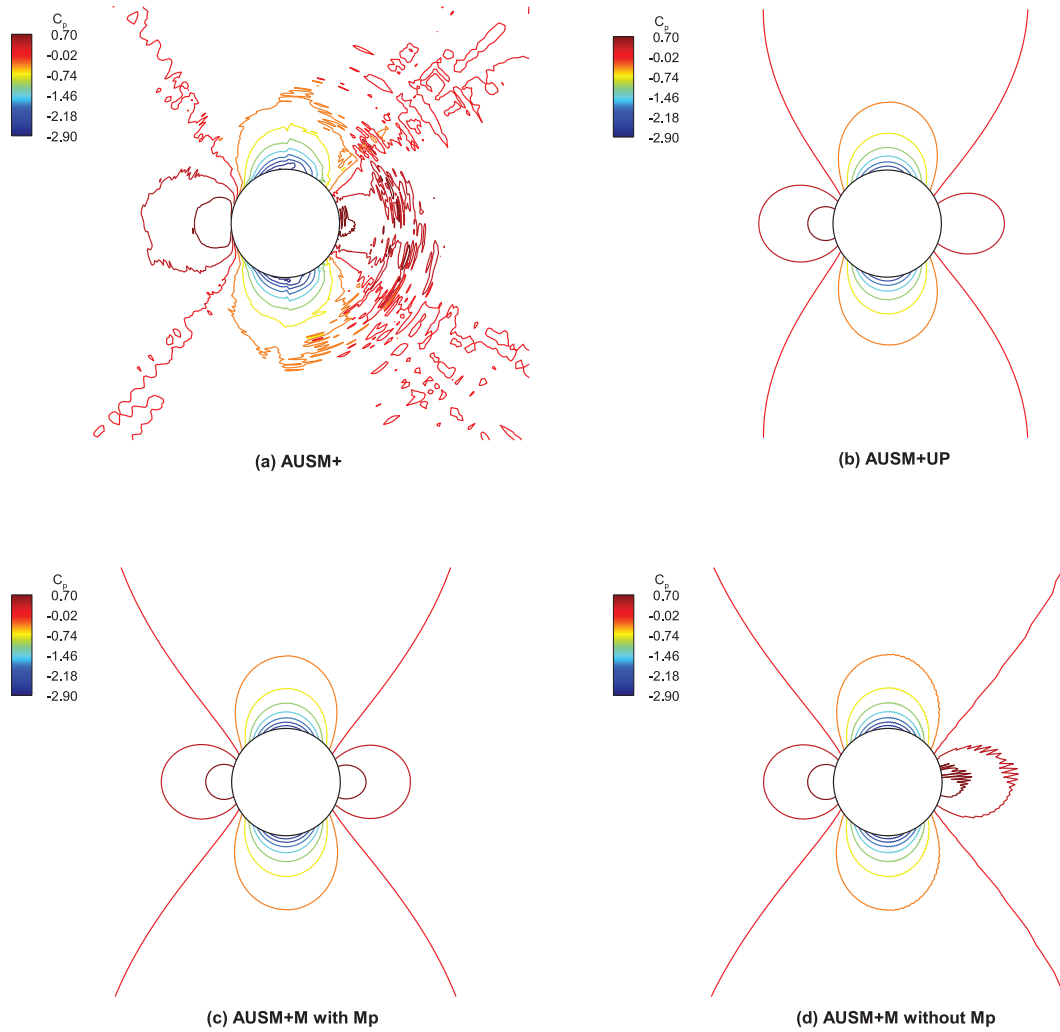


Fig. 2. Pressure coefficient contours for Mach=0.01 inviscid flow: (a) AUSM+; (b) AUSM+UP; (c) AUSM+M with M_p (Eq. (14)); (d) AUSM+M without M_p (Eq. (14)).

4.1. Effect of pressure diffusion term

Let us consider the effect of the pressure diffusion term M_p (Eq. (14)) on pressure oscillation. Inviscid flow over a cylinder at Mach=0.01 and hypersonic laminar flow over a cylinder at Mach=8.1 are chosen to illuminate the effectiveness of M_p . All computations are conducted by the implicit LU-SGS (Lower–Upper Symmetric Gauss–Seidel) approach [38] and second-order MUSCL (monotone upstream-centered schemes for conservation laws) approach reconstruction with van Albada limiter [39].

For Mach=0.01 inviscid flow, the mesh contains 301 (circumferential) $\times$ 401 (wall normal) grid points with the 20R exterior boundary. AUSM+ and AUSM+M adopt a large Courant–Friedrichs–Lewy (CFL) number (30,000 time iterations with CFL=100) to obtain the solution, but AUSM+UP diverges with CFL=100 due to the global cut-off problem. Such dilemma of AUSM+UP, which has been occasionally reported by [34], can be alleviated using the lower time step and higher cut-off reference Mach number [35,36]. In the current work, a small CFL number (400,000 time iterations with CFL=0.03) is applied in AUSM+UP. Fig. 2 displays the pressure coefficient contours computed by different numerical schemes. AUSM+ gives the very poor pressure distributions on account of deteriorated accuracy and checkerboard mode, and meantime AUSM+M (without M_p) still produces instable pressure oscillations. In contrast, AUSM+UP and AUSM+M (with M_p) obtains the smooth pressure distributions. In addition, AUSM+M allows the larger CFL number than AUSM+UP, since the improved pressure diffusion term M_p cancels the Mach number in the denominator and averts the global cut-off strategy.

For Mach=8.1 viscous flow, the flow conditions from [8] are set as: no angle of attack, $T_\infty = 63.73\text{K}$, $T_{\text{wall}} = 300\text{K}$, and $Re_\infty = 1.3 \times 10^5$ with respect to the radius $r = 20\text{mm}$. The mesh has 135 (circumferential) $\times$ 147 (wall normal) grid points clustered in the normal direction to shock wave. The cell Reynolds number is $Re_{\text{cell}} = 1.5$. The computations are done up to 50,000 time iterations with CFL=5. Fig. 3 plots the pressure contours for hypersonic viscous flow over a cylinder. In com-

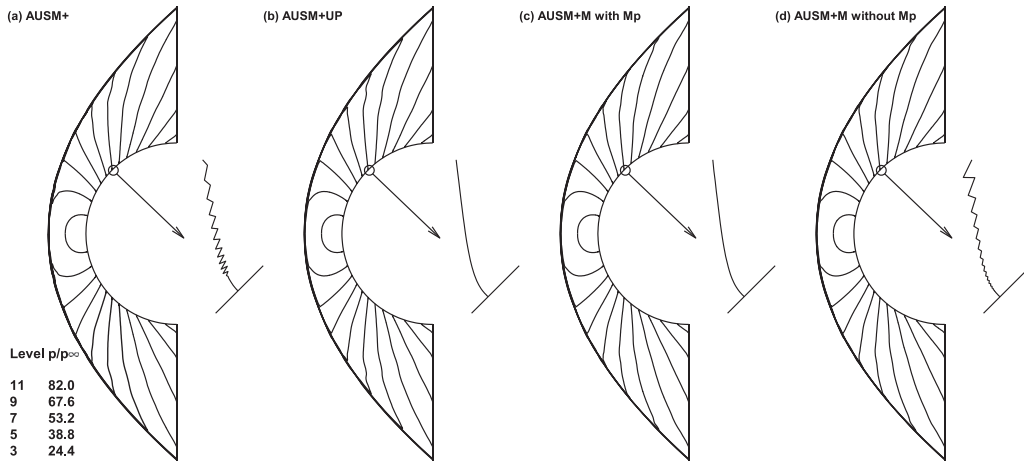


Fig. 3. Pressure contours (p/p_∞) for Mach=8.1 viscous flow: (a) AUSM+; (b) AUSM+UP; (c) AUSM+M with M_p ; (d) AUSM+M without M_p .

parison with AUSM+ and AUSM+M (without M_p) which produces pressure wiggles near the wall, AUSM+UP and AUSM+M (with M_p) eliminate this undesirable phenomenon. These two cases remarkably demonstrate that the pressure diffusion term M_p becomes significant to stabilize low-speed simulation and damp pressure wiggles near the wall for hypersonic simulation.

4.2. Effect of velocity diffusion term

We turn our attention to the effect of the velocity diffusion term (Eq. (26)) on shock robustness. The classic odd-even decoupling proposed by Quirk [9] is often applied to check shock stability of numerical scheme. For this case, the initial condition is given as:

$$(\rho, u, v, p)_L = (1, 6, 0, 1), (\rho, u, v, p)_R = (5.25, 0.353, 0, 40.64)$$

It yields a normal shock with Mach=6. The simulations are done by first-order accurate method and the explicit 3th TVD Runge-Kutta scheme with CFL=0.4. Many investigators [6,8,40] have revealed that the aspect ratio cells have a remarkable influence on shock robustness. Thus, to highlight the efficacy of Eq. (26), we generate the mesh with $200(X) \times 80(Y)$ grid cells in $[0, 10] \times [0, 1]$ with a large aspect ratio. The initial shock is located at $x=8$. The centerline grid is disturbed by small numerical error $\pm 10^{-4}$.

Fig. 4 provides density contours of the odd-even decoupling at $t=5$ s computed by AUSM+, AUSM+UP and AUSM+M schemes. Some important observations can be made from Fig. 4: (a) AUSM+, AUSM+UP and AUSM+M (without Eq. (26)) produce anomalous oscillations near shock wave with the current mesh, implying that they are prone to carbuncle phenomenon; (b) AUSM+M (with Eq. (26)) is free from shock anomalies and reproduces stable shock. This illustrates that the improved velocity diffusion term (Eq. (26)) strengthens shock robustness. The following computational results (Sections 5.1 and 5.4) will further show the superiority of AUSM+M (with Eq. (26)) compared with many existing upwind schemes.

4.3. Effect of numerical sound speed

Let us examine the effect of the numerical sound speed (Eq. (28)) on shock resolution. The stationary oblique shock and expansion shock cases are performed to evaluate the capability of Eq. (28). All computations are simulated by first-order accurate method and the explicit 3th TVD Runge-Kutta scheme with CFL=0.4. The computational domain is $[0, 1] \times [0, 0.1]$ with 100×10 grid cells. The computational results along horizontal centerline are displayed. It is reminded that four schemes SLAU, AUSM+, AUSM+UP, AUSM+M schemes are compared.

For stationary oblique shock, the initial condition is given by

$$(\rho, u, v, p)_L = (1, 1.5, 2.59808, 0.71429), (\rho, u, v, p)_R = (1.86207, 0.80556, 2.59808, 1.75595)$$

This generates 30° oblique shock with Mach=3. In Fig. 5, the pressure and Mach number distributions at $t=5$ s are displayed. SLAU, AUSM+, AUSM+UP and AUSM+M (with Eq. (12)) smear oblique shock even with shock-aligned mesh. AUSM+M (with Eq. (28)) precisely reproduces an oblique shock in one cell interface. The numerical sound speed (Eq. (28)) improves the resolution of oblique shock.

For expansion shock, the initial condition is specified as

$$(\rho, u, v, p)_L = (3.85714, 0.25926, 0, 0.820106), (\rho, u, v, p)_R = (1, 1, 0, 0.079365)$$

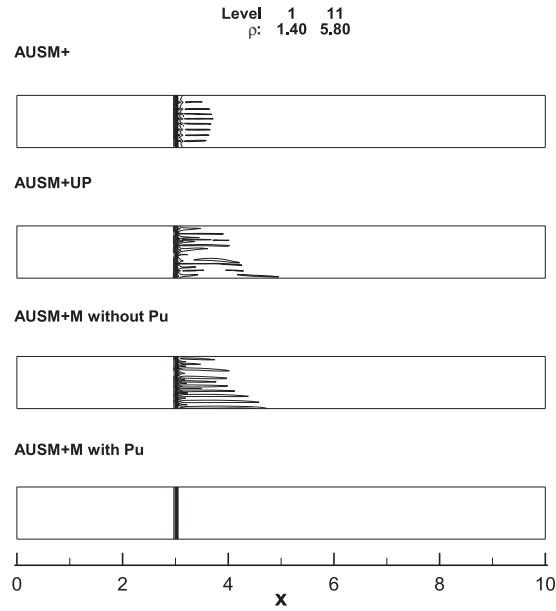


Fig. 4. Density contours of the odd-even decoupling.

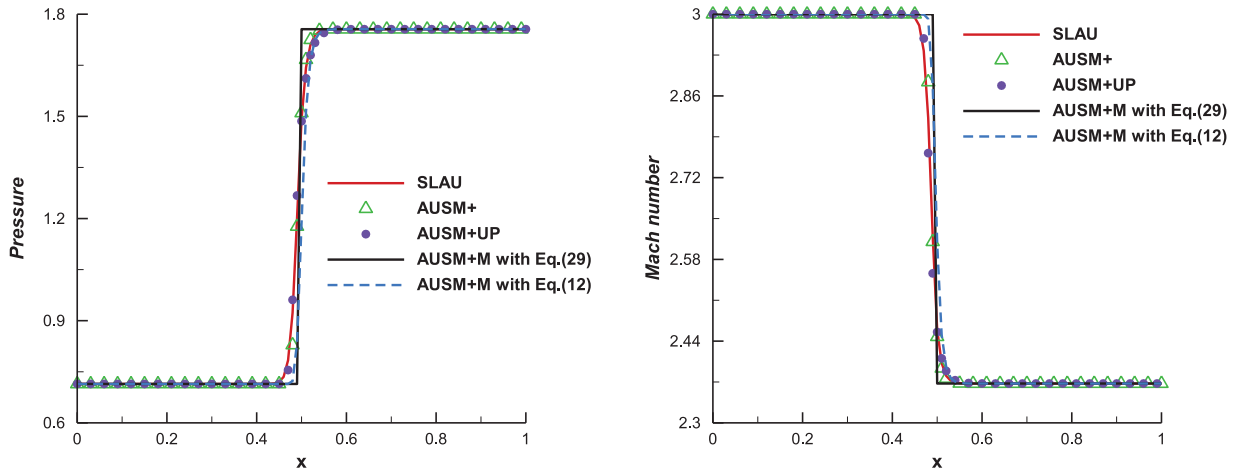


Fig. 5. Stationary oblique shock: (Left) Pressure; (Right) Mach number.

This condition at Mach=3 inverts the Rankine-Hugoniot relation [41]. Fig. 6 shows the density and Mach number distributions at $t=0.1$ s. As can be seen, AUSM+ predicts the unphysical entropy-violating expansion shock [18]. SLAU, AUSM+UP and AUSM+M remove this unphysical expansion shock and produce a smooth transition through sonic point. The numerical sound speed (Eq. (28)) matches the entropy-satisfying property.

4.4. Additional features based on shock tube

A variety of compressible shock tubes are calculated to present the additional features of AUSM+M. The length of shock tube is one with 100 grid cells. All simulations are done with first-order accurate method and the explicit 3th TVD Runge-Kutta scheme with CFL=0.4.

4.4.1. Moving contact discontinuity

This flow is considered to examine the resolution of contact discontinuity associated with boundary-layer resolution. The initial condition is set as:

$$(\rho, u, p)_L = (0.125, 0.1125, 1), (\rho, u, p)_R = (10, 0.1125, 1)$$

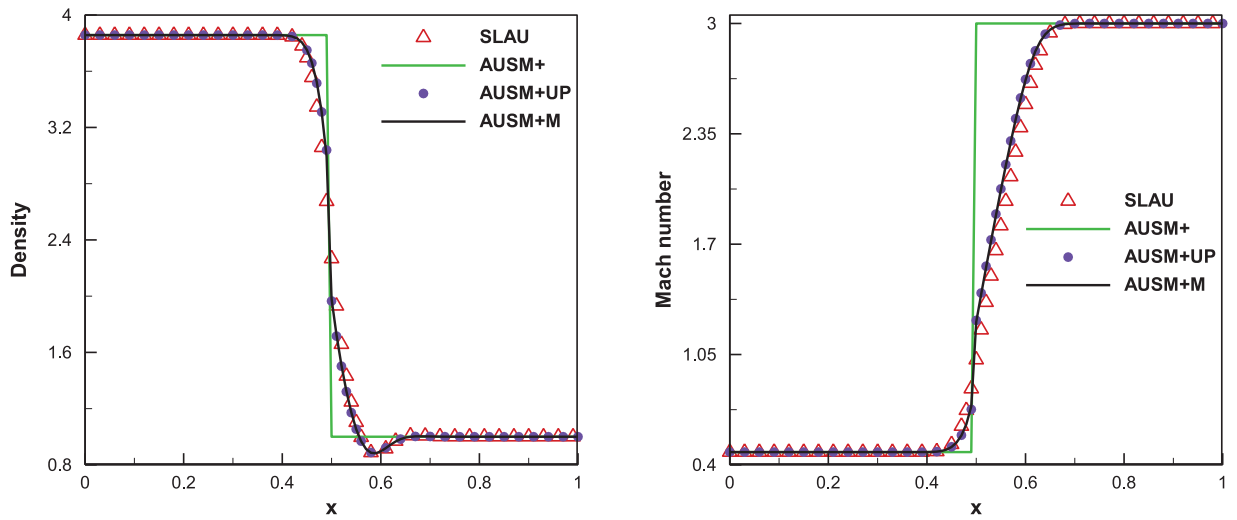


Fig. 6. Expansion shock: (Left) Density; (Right) Mach number.

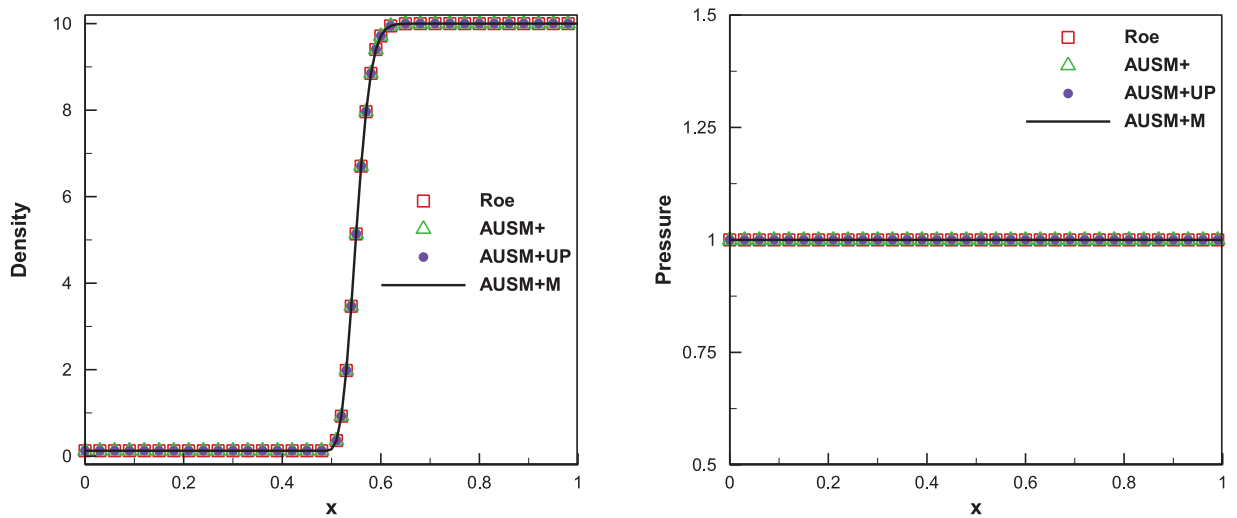


Fig. 7. Moving contact discontinuity: (Left) Density; (Right) Pressure.

In Fig. 7, we provide the density and pressure distributions at $t=0.5$ s computed by Roe, AUSM+, AUSM+UP and AUSM+M. The computational result of AUSM+M is identical with Roe, AUSM+ and AUSM+UP. AUSM+M is capable of capturing moving contact discontinuity exactly.

4.4.2. Positivity property

Now we concern the vacuum flow problem to calibrate the positivity and robustness of numerical flux. The initial condition is specified as

$$(\rho, u, p)_L = (1, -2, 0.4), (\rho, u, p)_R = (1, 2, 0.4)$$

This produces a drop of density and pressure in the middle. Fig. 8 displays the density and pressure distributions at $t=0.12$ s simulated by different numerical schemes. Roe scheme even with entropy fix turns out computational divergence [42]. For the computed results, there is almost no difference between AUSM+M and AUSM+, AUSM+UP. AUSM+M obtains the physical solution and preserves positivity [42].

4.4.3. Monotonicity property

The flow of two colliding shocks [43] is performed to verify the monotonicity property. The initial condition is given by

$$(\rho, u, p)_L = (1, 1, 0.1), (\rho, u, p)_R = (1, -1, 0.1)$$

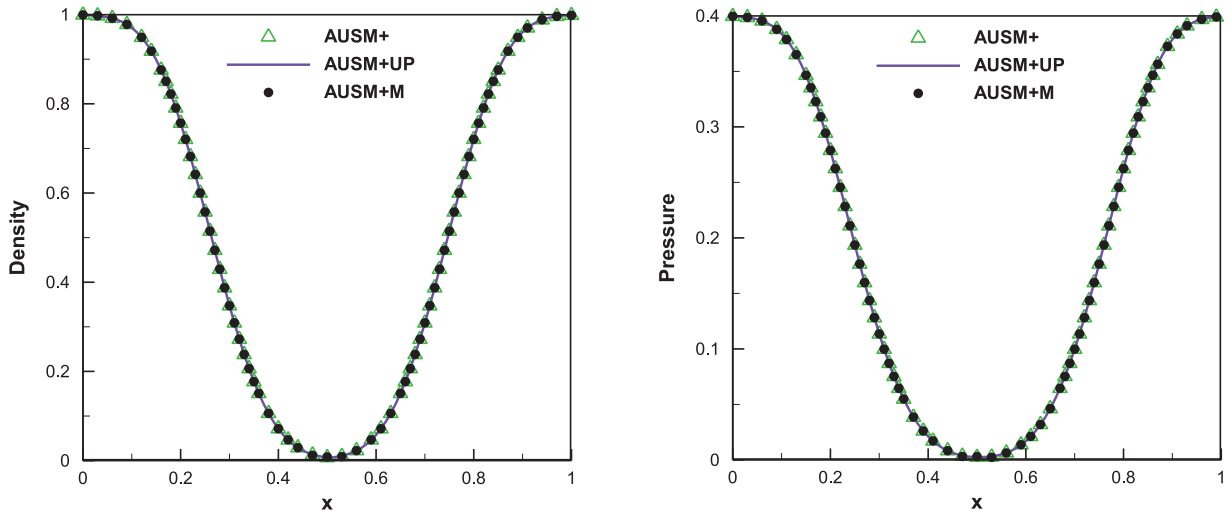


Fig. 8. Vacuum flow problem: (Left) Density; (Right) Pressure.

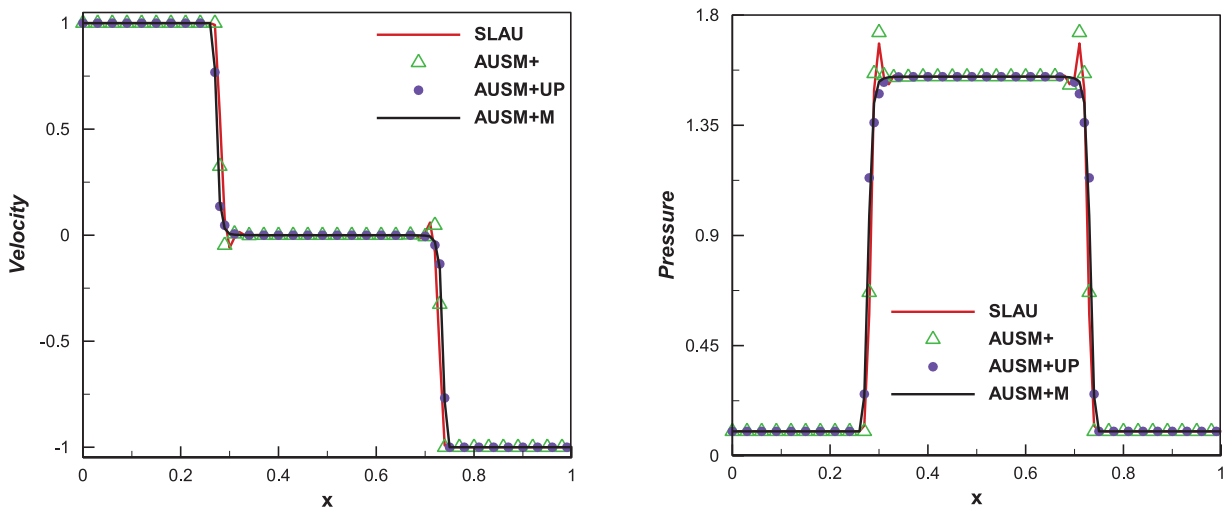


Fig. 9. Colliding shock: (Left) Velocity; (Right) Pressure.

The specific heat ratio is specified as $\gamma = 5/3$. In Fig. 9, we draw the velocity and pressure distributions at $t=0.5$ s computed by SLAU, AUSM+, AUSM+UP and AUSM+M. We can observe that AUSM+UP and AUSM+M obtain the monotonic profiles, whereas AUSM+ and SLAU produce numerical overshoots and slight oscillations at the shock. The proposed AUSM+M scheme damps numerical overshoots and satisfies monotonicity property.

5. Numerical results

Various numerical cases involving wide-ranging applications are carried out to highlight the effectiveness of the present AUSM+M scheme. All numerical experiments are conducted with the finite volume code based on multi-block structured mesh. The MUSCL reconstruction method is utilized to achieve higher-order extensions. The viscous flux is discretized with second-order central difference scheme. Other mature and reliable techniques are employed, including LU-SGS approach [38] for the implicit time integration.

5.1. Hypersonic inviscid flow over 2D cylinder

Comparative hypersonic flow over two-dimensional circular cylinder at Mach=20 is performed to evaluate the shock-capturing capability of AUSM+M scheme. This case is very challenging and a large number of literatures [1–3,6,9] have reported that many upwind schemes (e.g. Roe scheme) encounter the notorious carbuncle instability for hypersonic blunt body. The mesh (e.g. mesh geometry, mesh size and the aspect ratio cells) has an important influence on shock robustness.

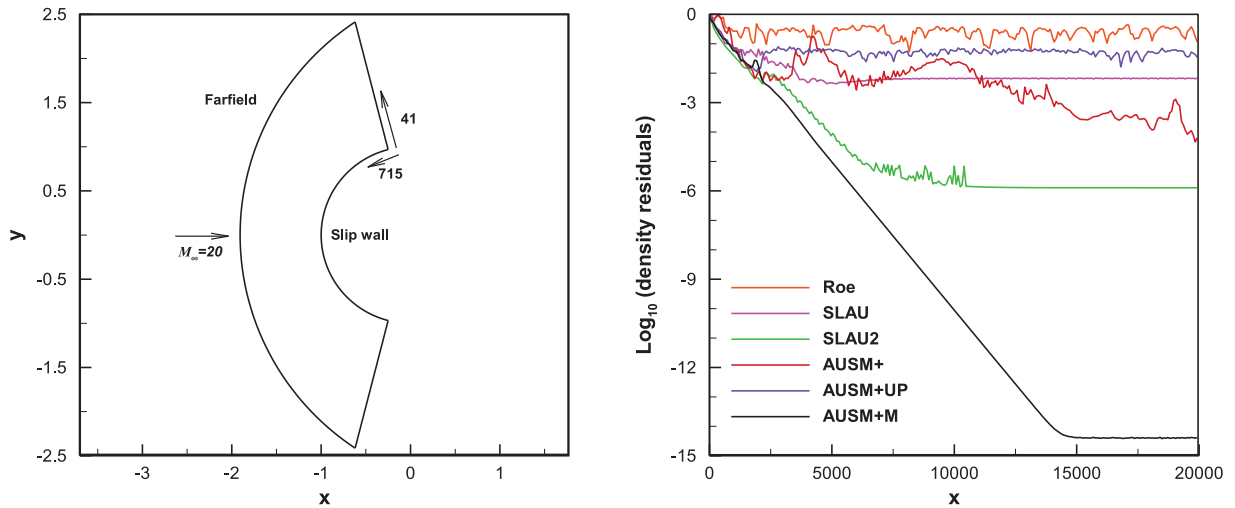


Fig. 10. Left: Cylinder mesh; Right: Convergence histories.

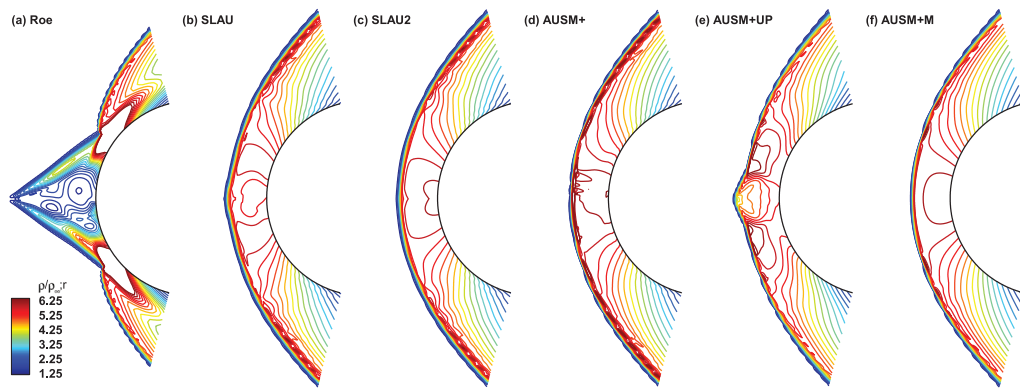


Fig. 11. Density contours (ρ/ρ_∞) for Mach=20 inviscid flow: (a) Roe; (b) SLAU; (c) SLAU2; (d) AUSM+; (e) AUSM+UP; (f) AUSM+M.

Many investigators [6,8,40] have demonstrated that large aspect ratio cells cause shock anomalies easily. Thus, to highlight the outstanding shock robustness of AUSM+M, the computational mesh with large aspect ratio cells is generated, consisting of 715 (circumferential) $\times$ 41 (wall normal) grid points in Fig. 10 (left). Steady-state flow simulations are done to 20,000 time iterations by first-order accurate method and LU-SGS with CFL=5.

Fig. 11 draws the density contours of the solutions using Roe, SLAU, SLAU2, AUSM+, AUSM+UP and AUSM+M schemes. Roe yields the most catastrophic carbuncle failings. Meanwhile, SLAU and AUSM+ is vulnerable to carbuncle instability under the large aspect ratio cells. AUSM+UP gives shock instability and SLAU2 produces slight anomalies. Free from such undesirable phenomenon, AUSM+M obtains smooth pressure distributions and predicts stable shock waves. Further, we plot convergence histories with different numerical schemes in Fig. 10 (right). Due to shock anomalies, the density residuals of Roe, SLAU, SLAU2, AUSM+ and AUSM+UP stagnate after about one to six orders of magnitude reductions. In comparison, AUSM+M attains the good convergence histories dropping to about 10^{-14} after 14,000 time iterations. It is confirmed that AUSM+M is more robust against shock anomalies than many existing schemes based on the current mesh.

5.2. Gresho vortex

To purely reveal the merit of AUSM+M at low Mach number flows, an unsteady inviscid Gresho vortex [44,45] is simulated. The formulation of the initial condition about Gresho vortex can be seen in [15,16,45], which allows different Mach numbers for the weakly compressible flow. This special vortex moves in the horizontal direction and keeps the vortex pattern without changing in time. The Mach number is selected as $M_0=0.03$. The computational mesh is a square of length 1, containing 121 grid points in each direction. A farfield boundary condition is imposed at the lower and upper edges, whereas a periodic boundary condition is assumed at the left and right edges. All simulations are done up to one period time by first-order accurate method and the explicit 3th TVD Runge-Kutta scheme. AUSM+UP and SLAU are provided with high resolution of low Mach number flows to calibrate AUSM+M.

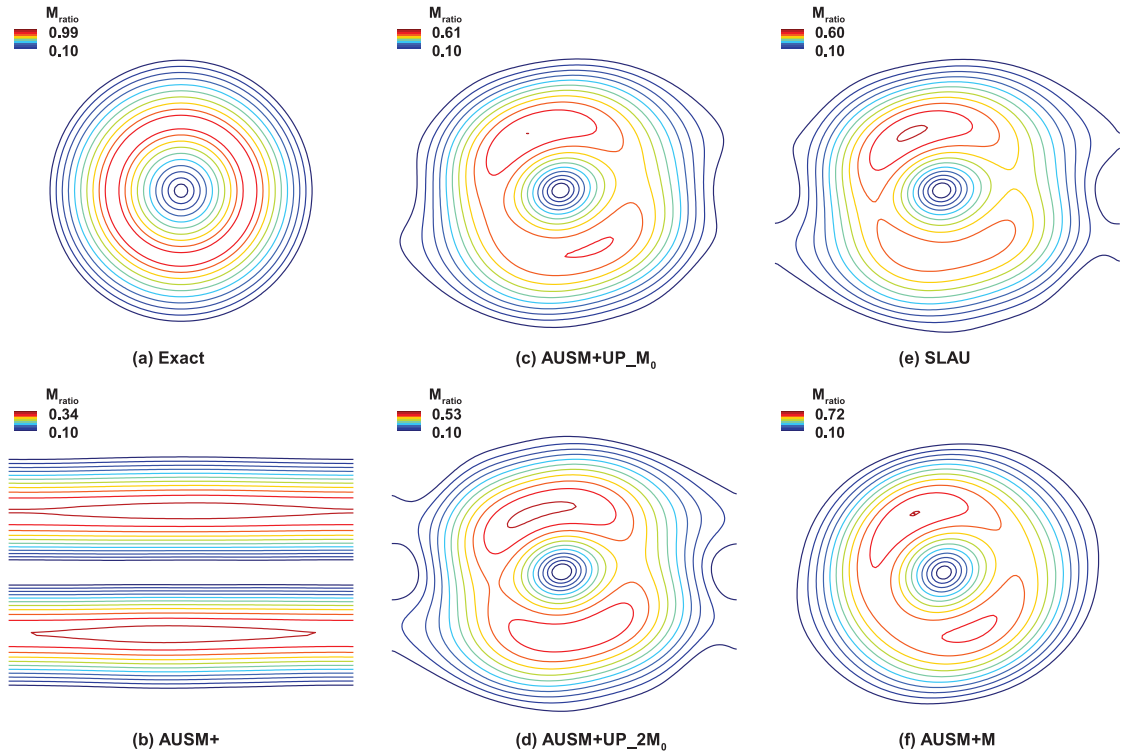


Fig. 12. Mach number ratio isolines for Gresho vortex.

The motivation to choose this case is twofold: firstly, to check the accuracy of AUSM+M in low Mach number. The exact solution of Gresho vortex is the initial value after one period. Such case can illustrate clearly how low the numerical viscosity of all-speed schemes is. Secondly, to investigate the robustness of AUSM+M at low speed. A time step of $\Delta t = 4e - 3$ is applied in AUSM+, SLAU and AUSM+M to obtain the solution. However, the computation of AUSM+UP is divergent with $\Delta t = 4e - 3$. To acquire the successful result, we reduce the time step or increase the reference Mach number to get over this difficulty of AUSM+UP [35,36]. The maximum time step based on the original reference Mach number of $M_\infty = 0.03$ is $\Delta t = 4e - 4$ for AUSM+UP. The increased reference Mach number of $M_\infty = 2M_0$ allows the maximum time step $\Delta t = 9e - 4$ for AUSM+UP. AUSM+M avoids such difficulty and permits the larger time step compared with AUSM+UP. This indicates that at low speeds, AUSM+M attains better robustness than AUSM+UP in terms of stability.

Fig. 12 displays the Mach number ratio isolines computed by AUSM+, AUSM+UP, SLAU and AUSM+M schemes. The Mach number ratio is defined as $M_{ratio} = \sqrt{[(u - u_0)^2 + (v)^2]/(\gamma p/\rho)/M_0}$. The maximum Mach number ratio can denote the level of numerical viscosity. For these numerical schemes, the maximum Mach number ratio is respectively 35.1% (AUSM+), 60.5% (SLAU), 61.1% (AUSM+UP with $M_\infty = M_0$), 53.5% (AUSM+UP with $M_\infty = 2M_0$) and 72.1% (AUSM+M). AUSM+ has the excessive numerical viscosity, which highly smears the computed vortex, whereas all-speed schemes (AUSM+UP, SLAU, AUSM+M) can capture this vortex with higher resolution. The differences are more visible in Fig. 13 in which pressure and Mach number distributions along the vertical centerline are drawn. AUSM+UP gives the more smeared distributions with the increased reference Mach number. In the current case, AUSM+M attains better accuracy than AUSM+, AUSM+UP and SLAU at low speeds.

5.3. Flat-plate laminar boundary layer

The flat-plate laminar boundary layer is considered to verify the shear-layer resolution of AUSM+M scheme. The flow conditions are given as follows: $Ma = 0.2$, the free stream Reynolds number $Re_\infty = 2.7 \times 10^4/m$. Shown in Fig. 14 (Left), the computational mesh has 367 (streamwise) $\times$ 181 (wall normal) grid points and the minimum mesh size near the wall is 1×10^{-2} mm. The lower boundary ($-500 < x < 0$) is symmetry and other lower boundary ($0 < x < 1500$) is no-slip wall. The second-order MUSCL with van Albada limiter is applied in this simulation.

We plot the velocity profiles at $x = 1000$ mm compared with the solution of Blasius in Fig. 14 (Right). The vertical ordinate η is defined as $\eta = y\sqrt{Re_x}/x$. The results of AUSM+UP and AUSM+M are identical and are consistent with the solution of Blasius. This demonstrates that AUSM+M has the boundary-layer-resolving capability.

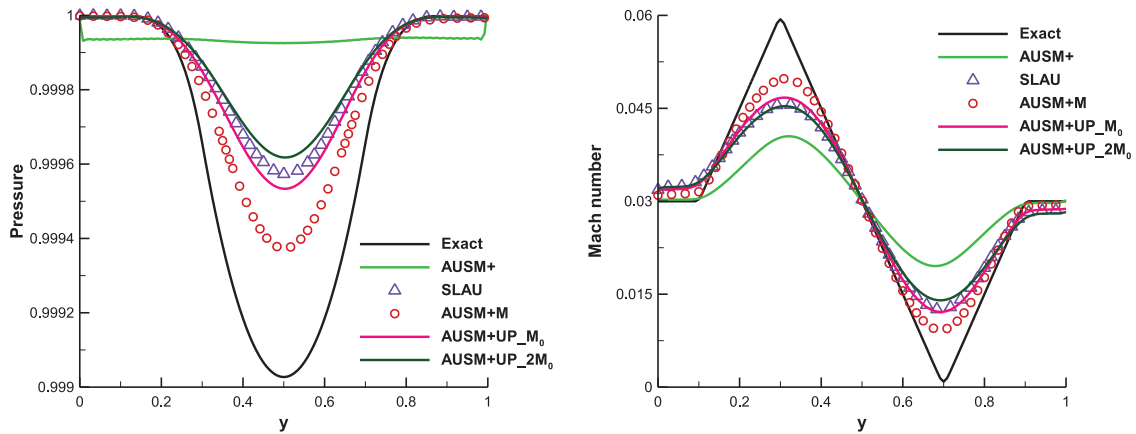


Fig. 13. Distributions along the vertical centerline of Gresho vortex: (Left) Pressure; (Right) Mach number.

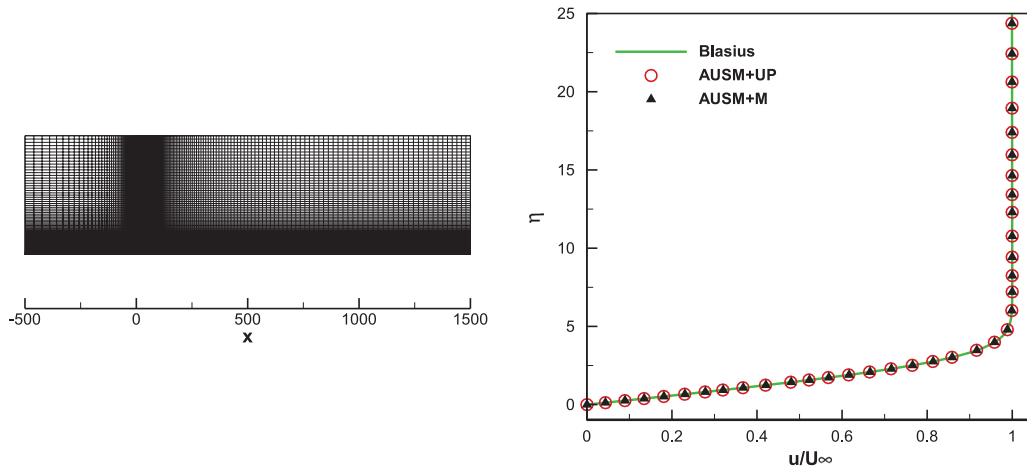


Fig. 14. Left: The mesh of flat plate; Right: Velocity profiles for flat plate.

5.4. Hypersonic viscous flow over 3D reentry capsule

Three-dimensional reentry capsule of Mars Science Laboratory (MSL) [46] is presented to show the advantage of AUSM+M in hypersonic complex viscous flows. Based on the Run3021 experimental condition [46], the freestream conditions are as follows: $M_\infty = 9.47$, $T_\infty = 54.8K$, $p_\infty = 167.9Pa$, $T_w = 294.44K$, $r = 76.2mm$, angle of attack $\alpha = 0^\circ$. The mesh of MSL forebody is shown in Fig. 15 (left). The total grid cells number is about 5.34×10^5 and the first grid distance in boundary layer is $5 \times 10^{-3}mm$ with the cell Reynolds number $Re_{cell} = 19.3$. We employ second-order accurate method (van Albada limiter) to simulate laminar flows. All computations are done up to 50,000 time iterations by LU-SGS with CFL=2.5. Here, we compare AUSM+M with AUSM+ and AUSM+UP.

In Fig. 15 (right), the symmetry plane is colored by Mach number contours. As can be seen, there is a bow shock wave close to the forebody of MSL as a result of hypersonic strong compressible effect. AUSM+ reproduces slight numerical oscillation and AUSM+UP obtains unphysical anomalous distributions for Mach number contours. The reason for this undesirable phenomenon is that the velocity diffusion term of AUSM+UP is based on one-dimensional characteristic relation and not suitable for the current multi-dimensional hypersonic vehicles. In comparison, AUSM+M circumvents these unphysical performances and gives smooth Mach number contours.

Fig. 16 provides the surface heating contours computed by three schemes. AUSM+UP gives the poor heat flux since it is still not robust enough against shock instability. AUSM+ presents the heat flux results with apparent oscillations. AUSM+M obtains the rational and smooth heat flux distributions. Fig. 17 draws the surface heating distributions along the centerline. Some observations can be drawn: (1) AUSM+UP shows the very poor heating profile with the wrong stagnation values; (2) AUSM+ is still not in good agreement with experimental data [46]; (3) AUSM+M attains the improvement of the peak heating value and the desirable heating profiles consistent with experimental one. This case indicates that the improved velocity diffusion term of AUSM+M is more suitable for hypersonic computations. AUSM+M achieves better robustness and accuracy for hypersonic heating prediction.

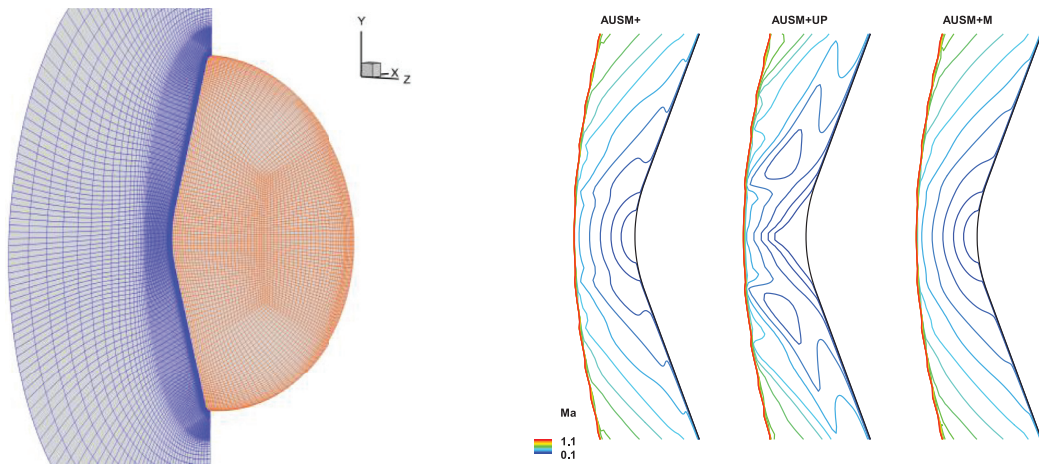


Fig. 15. Left: The mesh of MSL forebody; Right: Mach number contours of MSL on symmetry plane.

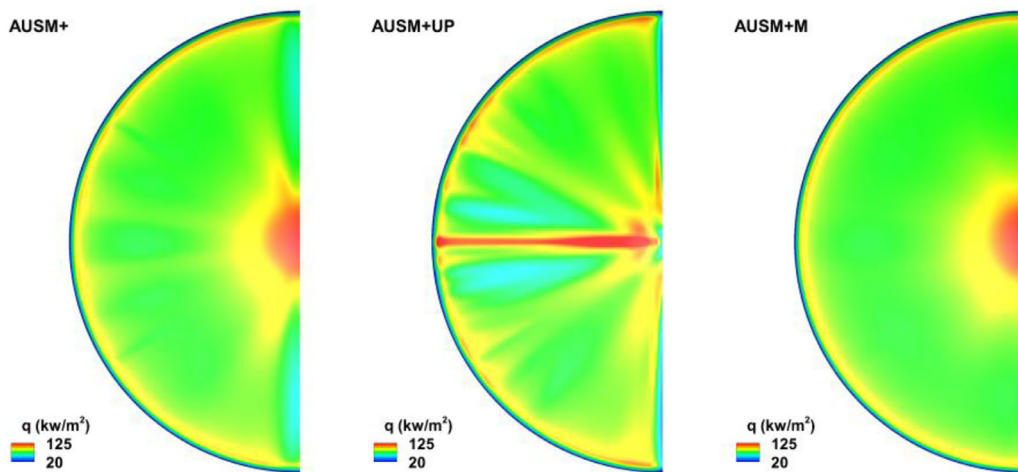


Fig. 16. The equally spaced heating contours for MSL.

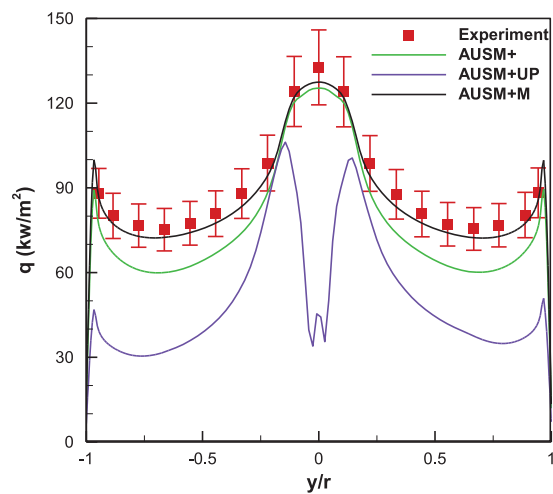


Fig. 17. The surface heating distributions along the centerline.

6. Conclusions

The paper has developed a newly improved AUSM-family scheme for all Mach number flows, called AUSM+M. The proposed scheme ameliorates pressure diffusion term, velocity diffusion term and numerical sound speed respectively to achieve more reliability in all-speed regimes. Series of benchmark test cases are performed to reveal that: (a) The improved pressure diffusion term is beneficial to stabilize low-speed simulation and suppress pressure wiggles near the wall for hypersonic simulation, and meantime it circumvents the global cut-off strategy and allows the larger time step compared with AUSM+UP; (b) The improved velocity diffusion term strengthens shock robustness to combat carbuncle instability; (c) The adopted numerical sound speed improves the resolution of oblique shock and matches the entropy-satisfying property; (d) AUSM+M also shares some favorable attributes, including resolving contact discontinuity exactly, damping numerical overshoots and positivity property. Hypersonic inviscid flow over a cylinder is further confirmed that AUSM+M is very robust against shock anomalies. Gresho vortex demonstrates that AUSM+M obtains the desired accuracy and averts the global cut-off strategy. Hypersonic viscous flow over 3D reentry capsule stresses the potential application of AUSM+M in hypersonic heating prediction. All in all, AUSM+M is appealing with high robustness, accuracy, reliability for all speed flows. It is worth further studying for complex flows.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.apm.2019.09.005](https://doi.org/10.1016/j.apm.2019.09.005).

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